

PHY 211

Lecture 6 Notes

By

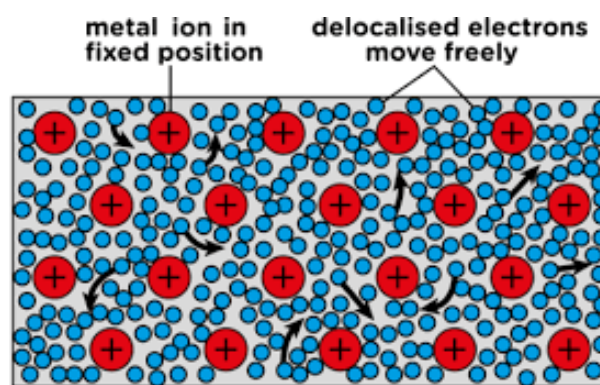
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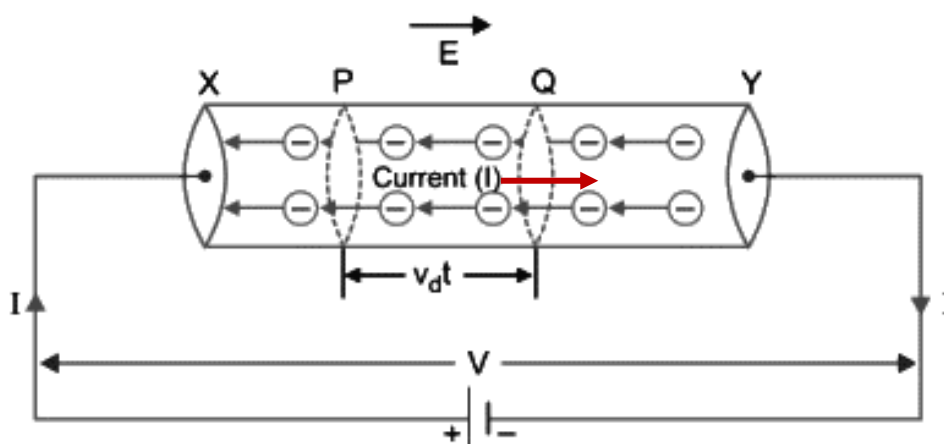
Chapter 5

Electric Current

- For uncharged metallic conductor at rest, the thermal internal energy of the material is sufficient to liberate the outer e_s^- of atoms.
- These e_s^- are called “free e_s^- ” because they are free to move through the material.
- Due to random motion of free e_s^- , the net flow of charge at any point is zero (i.e. **Zero current**).

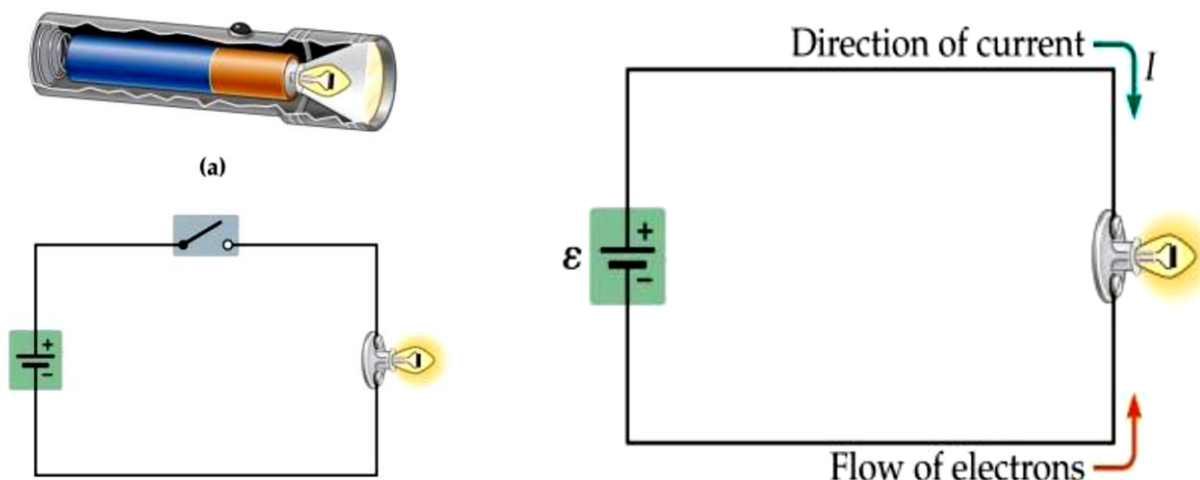


- Under the effect of the electric field and potential difference, free electrons in metals start to move in one direction with a velocity called **Drift velocity** and form electric current.



➤ The electric current “I”:

It is the net amount of charge that passes through a given cross-section of the conductor per unit time



- The electrons move from $-ve$ terminal to $+ve$ terminal.
- The conventional current is the current due to flow of $+ve$ charges (from $+ve$ terminal to $-ve$ terminal)

$$I = \frac{Q}{t} \text{ or in general } I = \frac{\Delta Q}{\Delta t} \text{ or } I = \frac{dQ}{dt} \quad \left(\text{Unit: } \frac{C}{s} = A \text{ "Ampere"} \right)$$

The total charge is given by:

$$Q = \text{no. of electrons} \times \text{electronic charge}$$

$$\therefore Q = Ne$$

• Example 1:

A steady current of 2.5 A flows in a wire for 4 minutes

- a) How many charges pass through any point in the circuit?
- b) How many electrons would that be?

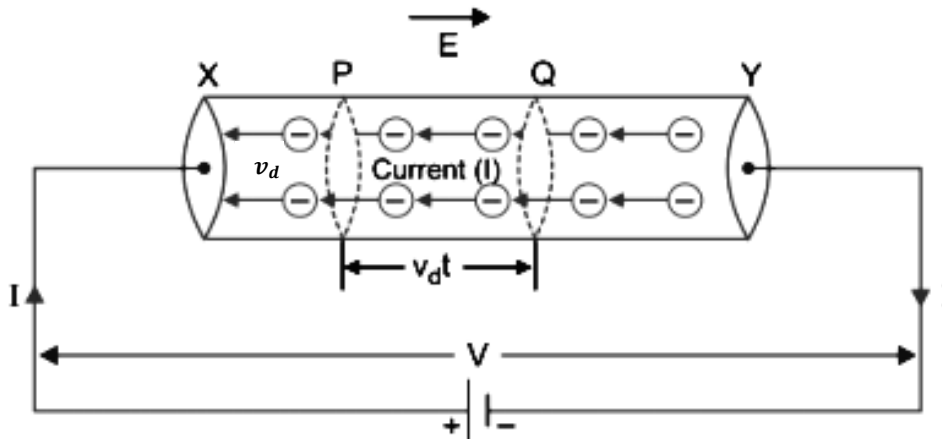
Solution

$$a) Q = It = 2.5 \times (4 \times 60) = 600 \text{ C}$$

$$b) N = \frac{Q}{e} = \frac{600}{1.6 \times 10^{-19}} = 3.8 \times 10^{21} \text{ electrons}$$

➤ **Drift velocity " v_d ":**

"It is the average velocity that free electrons get drifted with towards +ve terminal when electric field E is applied"

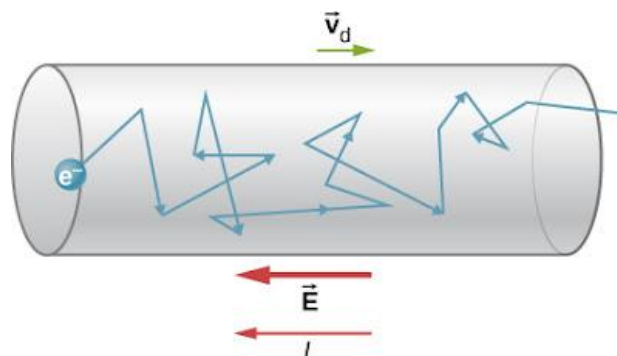
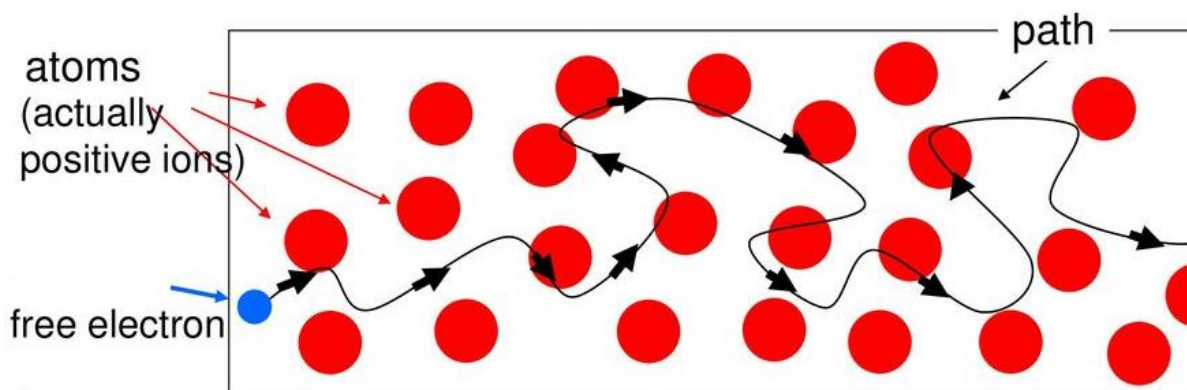


$$\vec{v}_d \propto \vec{E} \rightarrow \boxed{\vec{v}_d = \mu \vec{E}}$$

Mobility " μ ": the drift velocity per unit electric field $\left(\mu = \frac{v_d}{E}\right)$

$$\boxed{\mu = \frac{e\tau}{m}} \quad (\text{unit: } \text{m}^2\text{V}^{-1}\text{sec}^{-1})$$

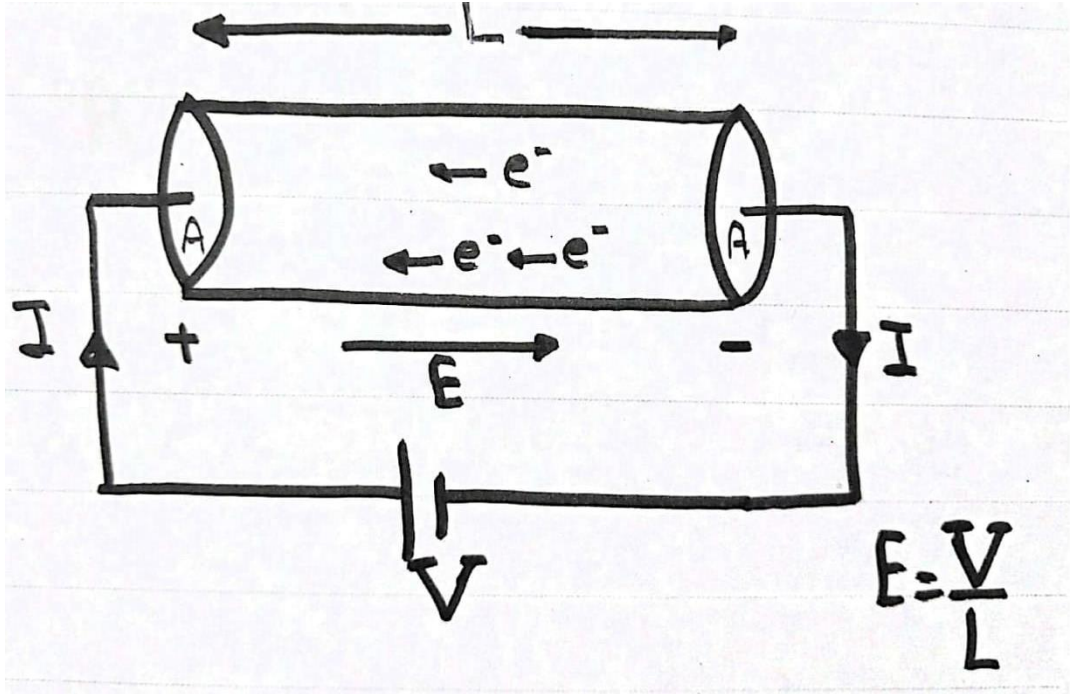
Relaxation time " τ ": the average time between two successive collisions.



➤ Current density " J ": "Current per unit area"

$$J = \frac{I}{A} \quad (\text{unit: } \text{A} \cdot \text{m}^{-2})$$

➤ Relation between current and drift velocity:



Charge density " n ": no. of charges " N " per unit volume (concentration of charge)

$$n = \frac{N}{V} = \frac{N}{AL}$$

$$I \propto v_d \quad \& \quad I \propto n \quad \& \quad I \propto A$$

$$I = nev_d A$$

$$J = \frac{I}{A} = nev_d$$

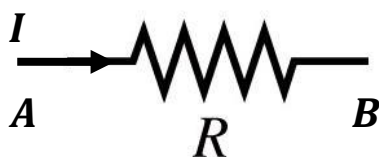
➤ Ohm's Law:

The current that flows through a conductor is directly proportional to the potential difference applied to its ends.

$$V \propto I \longrightarrow V = IR$$

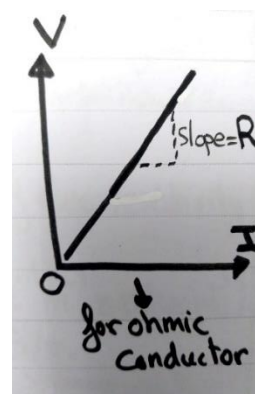
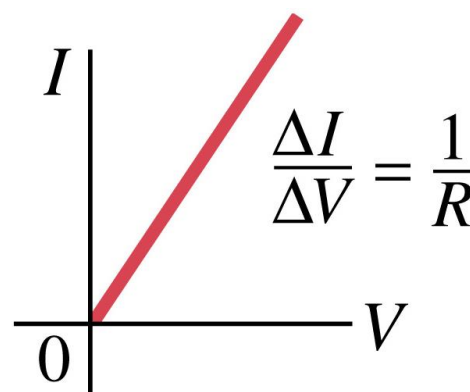
R: resistance of the conductor to the flow of current

$$R = \frac{V}{I} \left(\frac{V}{A} = \Omega \text{ "Ohm"} \right)$$

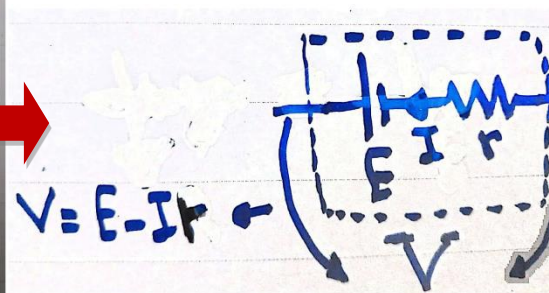
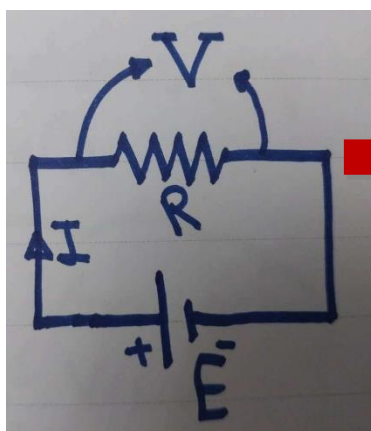


$$V_A - V_B = IR$$

$$V_B - V_A = -IR$$



➤ Internal resistance of battery "r":



$$V = IR$$

$$(emf) E = V + Ir, \quad r: \text{internal resistance of battery}$$

$$V = E - Ir \quad \& \quad r = \frac{E - V}{I} = \frac{E - V}{V} R$$

If the battery is not connected to a load:

$$\text{Terminal voltage } V = E \text{ (emf)}$$

If the battery is not connected to a load:

$$\text{Terminal voltage } V = E - Ir$$

• **Example 2:**

A small flashlight bulb draws 300 mA from its 1.5 V battery.

a) What is the resistance of the bulb?

b) If the voltage is dropped to 1.2 V, how would the current change?

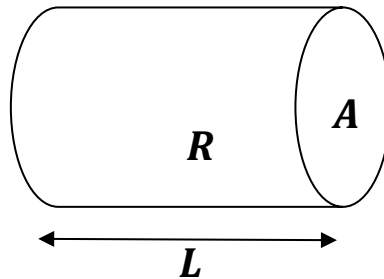
Solution

$$a) R = \frac{V}{I} = \frac{1.5}{300 \times 10^{-3}} = 5 \Omega$$

b) $\because$ the resistance of remains constant

$$\therefore I = \frac{V}{R} = \frac{1.2}{5} = 0.24 \text{ A}$$

➤ **Resistivity "ρ":**



The resistivity of a conductor is an internal property defined as

$$\rho = \frac{RA}{L} \quad (\Omega \cdot m)$$

A: area of conductor

L: length of conductor

ρ is constant for the conductor at constant temp so does not change with A or L.

$$\boxed{R = \frac{\rho L}{A}} \rightarrow R \propto L \quad \& \quad R \propto \frac{1}{A}$$

➤ **Conductivity "σ":** reciprocal of electrical resistivity.

$$\boxed{\sigma = \frac{1}{\rho} = \frac{L}{RA}} \quad \text{Unit: } \Omega^{-1} \text{m}^{-1} \text{ or } \text{S} \cdot \text{m}^{-1} \text{ "S: Siemens"}$$

• **Example 3:**

Given that the resistivity of a copper wire is $1.7 \times 10^{-8} \Omega \cdot \text{m}$. Find:

- The diameter of a 20 m wire if the resistance of the wire is 0.1Ω
- What is the voltage drop across the wire if the current flowing through the wire is 12 A?

Solution

$$a) A = \frac{\rho l}{R} = \frac{(1.7 \times 10^{-8}) \times 20}{0.1} = 3.4 \times 10^{-6} \text{ m}^2$$

$$\therefore A = \pi r^2 \longrightarrow r = \sqrt{\frac{A}{\pi}}$$

$$\therefore d = 2r = 2 \sqrt{\frac{A}{\pi}} = \sqrt{\frac{3.4 \times 10^{-6}}{\pi}} = 2.1 \times 10^{-3} \text{ m} = 2.1 \text{ mm}$$

$$b) V = IR = 12 \times 0.1 = 1.2 \text{ V}$$

➤ **Temperature effect on resistivity:**

The resistivity of a conductor is related to temp. as given

$$\boxed{\rho_T = \rho_o(1 + \alpha T)}$$

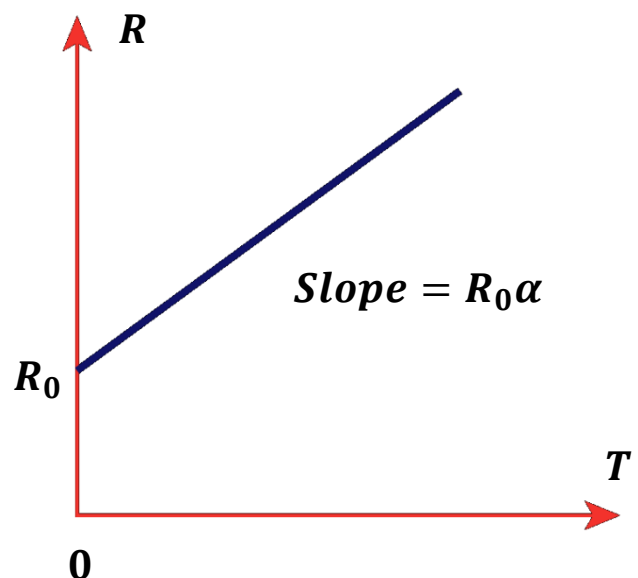
α : coefficient of increase in resistivity with temp. (unit: $^{\circ}\text{C}^{-1}$ or K^{-1})

ρ_o : resistivity at zero temperature.

ρ_T : resistivity at temperature T.

Similarly:

$$\boxed{R_T = R_o(1 + \alpha T)}$$



➤ Electric Energy “U” & Power “P”

When a small amount of charge ΔQ moves across a potential difference V , its electrical potential energy U changes by the amount

$$\boxed{\Delta U = V\Delta Q} \text{ (Joule)}$$

$$\boxed{\Delta U = IVt}$$

➤ Power: rate of change of energy

$$\boxed{P = \frac{\text{electric potential energy}}{\text{time}} \quad \left(\text{unit: Watt} = \frac{\text{J}}{\text{sec}}\right)}$$

$$\boxed{\therefore P = \frac{\Delta U}{t} = \frac{V\Delta Q}{t} = VI}$$

$$\boxed{\therefore P = V\left(\frac{V}{R}\right) = \frac{V^2}{R}}$$

$$\boxed{\therefore P = (IR)I = I^2R}$$

- **Example 4:** Calculate the resistance of 40 W automobile head-light designed for 12 V.

Solution

$$P = \frac{V^2}{R} \longrightarrow R = \frac{V^2}{P} = \frac{12^2}{40} = 3.6 \, \Omega$$

➤ Electric bills:

In electrical appliances, larger units such a kilowatt-hour (kWh) may be used for energy.

$$\text{Energy} = \text{Power} \times \text{time}$$

$$\boxed{1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ (W} \cdot \text{sec} = \text{J})}$$

- **Example 5:**

An electric heater draws 15.0 A on a 240 V line. How much power does it use and how much does it cost per month (30 days) if it operates 3 hours per day and the rate is 10.5 cents per kWh. Assume the current flows steadily in one direction.

Solution

The power consumed is

$$P = IV = 15 \times 240 = 3600 \text{ W} = 3.6 \text{ kW}$$

Hours of usage are

$$(3 \text{ hours/day})(30 \text{ days}) = 90 \text{ hours}$$

The electrical energy usage is

$$3.6 \times 90 = 324 \text{ kWh}$$

Therefore, the cost is

$$324 \text{ kWh} \times 0.105 \text{ \$/kWh} = \$34.02$$

- **Example 6:**

A 75 W light bulb operates on a potential difference of 95 V. Find

- a) The current in the bulb
- b) The resistance of the bulb
- c) If this bulb is replaced with one whose resistance is half the value of part (b), is its power rating greater than or less than 75 W? By what factor?

Solution

$$a) I = \frac{P}{V} = \frac{75}{95} = 0.79 \text{ A}$$

$$b) R = \frac{V}{I} = \frac{95}{0.79} = 120 \text{ } \Omega$$

$$c) R = \frac{1}{2} \times 120 = 60 \text{ } \Omega$$

$$P = \frac{V^2}{R} = \frac{95^2}{60} = 150 \text{ (greater)} \rightarrow \text{factor} = \frac{150}{75} = 2$$

- **Example 7:**

The current in a 120 V reading lamp is 2.3 A. If the cost of electrical energy is \$ 0.075 per kilowatt-hour, how much does it cost to operate the light for an hour?

Solution

$$U = P \times t = IVt = 2.3 \times 120 \times (1 \text{ hr}) = 276 \text{ Wh} = 0.276 \text{ kWh}$$

$$\text{Cost} = 0.276 \times 0.075 = \$0.021$$